

2.5.2 Non-Linear Data Structures (Tree, Graph)

ٹری ڈیٹا کو درجہ بندی (Hierarchy) میں رکھتی ہے جو ایک جڑ (Root) سے پھوٹتی ہے۔

CS XII

Chapter 2: Computational Thinking & Algorithms

Lecture 13 of 16

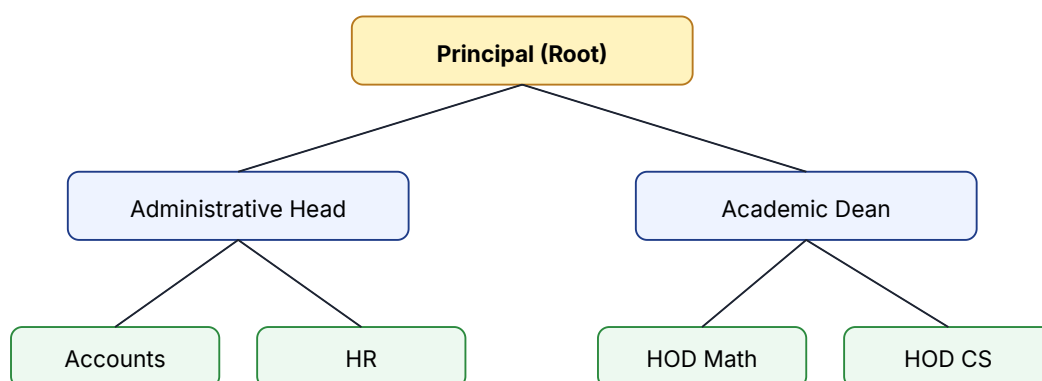
2.5.2 Non-Linear Data Structures (Tree, Graph)

LEARN IT

A **tree** organises data in a **hierarchy**. Everything branches out from a single starting point called the **root**.

It creates a strict **one-to-many (parent-child)** relationship. That is an excellent way to show how roles flow in an organisation.

Real-life examples: a family tree (grandparents → parents → children) and folders inside folders on a drive.



College tree: one root (Principal), then parent-child branches. No child has two parents in this model.

WORKED EXAMPLE

Worked example: name the relationships.

- Root: Principal.
- Children of the root: Administrative Head, Academic Dean.
- Children of Academic Dean: HOD Math, HOD CS.
- HOD CS has one parent (Academic Dean), not two — that is the tree rule. If HOD CS also reported sideways to Accounts as an equal link, the picture would become a **graph**.

REMEMBER IT

Tree = family / folders / organogram. One root. Branches down. Parent-child, not a web of equals.

COMMON MISTAKE

Mistake: drawing a tree with two roots, or with an arrow going back up to the Principal from HOD CS (a cycle).

Fix: one root, links only parent → child, no loops. Cycles belong to graphs.

ٹری ڈیٹا کو درجہ بندی (Hierarchy) میں رکھتی ہے جو ایک جڑ (Root) سے پھوٹتی ہے۔ تعلق والدین-بچہ (ایک سے کئی) ہوتا ہے، جیسے پرنسپل → ڈین → ایچ او ڈی، خاندانی شجرہ، یا فولڈر کے اندر فولڈر۔ دو جڑیں یا چکر ٹری کو گراف بنا دیتے ہیں۔

CHECK YOURSELF

1. What is the starting node of a tree called?
2. What kind of relationship does a tree model?
3. Give two real-life tree examples from the source.

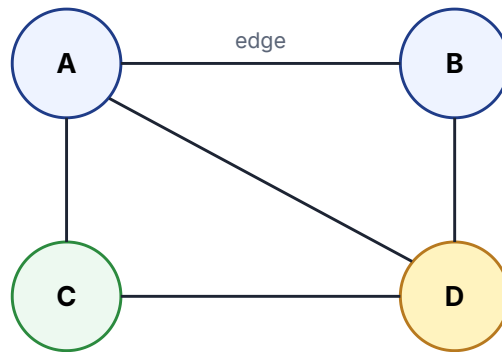
Answers: 1. The root. 2. One-to-many parent-child. 3. A family tree, and folders inside folders (or a college organogram).

LEARN IT

A **graph** organises data when things connect in a flexible, complex way. Unlike a tree, there is **no strict hierarchy**: any node can connect to any other node.

- **Nodes (vertices)** — the items (cities, people).
- **Edges** — the connections (roads, friendships).

Examples: Facebook friends; GPS mapping (the fastest route between two cities).



Four vertices A, B, C, D. Edges can form a cycle (A-B-D-C-A). A tree cannot do that.

WORKED EXAMPLE

Worked example: Karachi – Hyderabad – Sukkur – Larkana. Cities are vertices. Roads are edges. You can travel Karachi → Hyderabad → Sukkur or Karachi → Larkana → Sukkur. There is no single “root city”. That is a graph, and GPS uses it to pick a fast route.

REMEMBER IT

Tree $\subset$ Graph in shape-talk: a tree is a special graph with one root and no cycles. If you need cycles or many-to-many links, say **graph**.

COMMON MISTAKE

Mistake: using a tree for GPS because “Hyderabad comes under Sindh”.

Fix: administration can be a tree; **roads** are a graph because you can loop and take alternative paths.

گراف لچکدار روابط کے لیے ہے: کوئی سخت درجہ بندی نہیں، کوئی بھی نوڈ کسی سے جڑ سکتا ہے۔ نوڈز (Vertices) چیزیں ہیں، کنارے (Edges) تعلق۔ فیس بک دوست اور GPS نقشہ کلاسیکی مثالیں ہیں۔ چکر (Cycle) اور کئی راستے گراف کو ٹری سے الگ کرتے ہیں۔

CHECK YOURSELF

1. What are the two parts of a graph?
2. Give two source examples of graphs.
3. State one difference between a tree and a graph.

Answers: 1. Nodes/vertices and edges. 2. Social networks (Facebook friends) and GPS mapping. 3. A tree has a hierarchy and one root; a graph has no strict hierarchy and any node may connect to any other (cycles allowed).

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. An array stores items:

- a. In contiguous indexed slots of one type
- b. With arbitrary node links only
- c. As a hierarchy always
- d. As a pie

MCQ 2. A linked list stores items:

- a. In nodes with pointers to the next
- b. In one block of equal indexes only
- c. As a tree of bosses
- d. As a hash of pies

MCQ 3. A stack is:

- a. LIFO
- b. FIFO
- c. A graph
- d. A GUI

MCQ 4. A queue is:

- a. FIFO
- b. LIFO
- c. A binary tree
- d. A K-map

MCQ 5. Undo in an editor uses a:

- a. Stack
- b. Queue
- c. Pie
- d. OSI layer

MCQ 6. A printer line is a:

- a. Queue
- b. Stack
- c. AVL tree always
- d. NAND

Extra short questions

1. Define array, linked list, stack, queue in one line each.
2. One op each: push/pop, enqueue/dequeue.
3. Array vs list for insert at front.
4. Peek means:

Extra long question

1. Draw stack and queue after the sequence: insert A,B,C then remove one. Show remaining order and a real example of each. ★ Golden for stack/queue.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Array: indexed block. List: nodes+links. Stack: LIFO. Queue: FIFO.
2. Stack push/pop top. Queue enqueue rear, dequeue front.
3. Array shifts $O(n)$; list pointer change $O(1)$ if you have the head.
4. Read the top/front without removing.

Long

1. Stack remaining B,A (C popped). Queue remaining B,C (A served). Examples: back button vs ticket counter. Diagrams with top/front labelled.